

Functional Analysis — Homework 2

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Problem 1. Recall that c_0 is the space of sequences converging to zero, equipped with the supremum norm: $\|x\|_\infty = \sup_j |x_j|$. Consider the subspace:

$$S = \left\{ (x_1, x_2, x_3, \dots) : \sum_{j=1}^{\infty} \frac{x_j}{2^j} = 0 \right\}.$$

Show that there is no unit vector $x \in c_0$ with $d(x, S) = 1$. Hint: start with an arbitrary unit vector $x \in c_0$, and try to construct a vector $y \in S$ such that the distance $\|x - y\|_\infty$ is less than 1.

Problem 2. Let V be a normed space, and $x \neq y$ elements of V . Show that there exists a bounded linear functional $f \in V^*$ such that $f(x) \neq f(y)$.

Problem 3. Let V be a normed space, V^* its dual and V^{**} its bidual. Consider the map

$$j : V \rightarrow V^{**}$$

given by $j(v)(f) = f(v)$.

- (a) Show that j is well-defined, i.e. $j(v)$ is a bounded linear functional on V^* .
- (b) Show that j is linear.
- (c) Show that j is an isometry onto its image.

This problem shows that V embeds isometrically into a subspace of V^{**} under j . In general, j need not be surjective. When it is, we say that V is reflexive.

Proof.

- (a) For linearity of $j(v)$,

$$\begin{aligned} j(v)(af + bg) &= (af + bg)(v) \\ &= af(v) + bg(v) \\ &= aj(v)(f) + bj(v)(g). \end{aligned}$$

Now assume that $\|f\|_{op} = 1$. Then

$$\begin{aligned} \|j(v)(f)\| &= \|f(v)\| \\ &\leq \|f\|_{op} \|v\| \\ &= \|v\|, \end{aligned}$$

so since $\|v\|$ is constant as we vary f , $j(v)$ is bounded.

(b) We have

$$\begin{aligned}
 j(av + bx)(f) &= f(av + bw) \\
 &= af(v) + bf(w) \\
 &= aj(v)(f) + bj(w)(f) \\
 &= (aj(v) + bj(w))(f),
 \end{aligned}$$

where the second equality comes from the linearity of f .

(c) We already saw in the proof of part (a) that $\|j(v)(f)\| \leq \|v\|$ whenever $\|f\|_{op} = 1$. Thus $\|j(v)\|_{op} \leq \|v\|$, so if we can find an f with $\|f\|_{op} = 1$ such that $\|j(v)(f)\| = \|v\|$, we're done since we will have demonstrated that the supremum of $\|v\|$ is achieved. We can define a bounded linear map $f : \mathbb{R}v \rightarrow F$ by choosing to map $v \mapsto \|v\|$, and then by Hahn-Banach we can extend this to a bounded linear map $\tilde{f} : V \rightarrow F$ so that $\tilde{f} \in V^*$. Now we just check that indeed

$$\|j(v)(\tilde{f})\| = \|f(v)\| = \|v\|$$

as desired, and we are done. □

Problem 4. Let V be a normed space and

$$S_1(V) = \{v \in V : \|v\| = 1\}$$

be its unit sphere. Prove that V is separable if and only if $S_1(V)$ is separable.

Problem 5. (Uniform Boundedness Principle). Let X be a Banach space, Y a normed space, and $\{T_\alpha\}_{\alpha \in J}$ an arbitrary family of bounded linear operators

$$T_\alpha : X \rightarrow Y$$

satisfying the **pointwise bound**:

$$\text{For each } x \in X, \quad \sup_{\alpha \in J} \|T_\alpha(x)\|_Y < \infty.$$

Use the Baire Category Theorem to show that this implies the **uniform bound**:

$$\sup_{\alpha \in J} \|T_\alpha\|_{op} < \infty.$$